Group II, Row IV from the amplitude: the BFKL term

We start from your amplitude-level expression (with $\mathfrak{C}$ and $\mathfrak{A}^{(1)}$), do every step, and extract the log $\frac{\vee}{k^+}$ term. Only BFKL is discussed. The result is the same as before:

$$\boxed{\mathrm{BFKL}\big|_{\mathrm{II,IV}} = \log\frac{\vee}{k^+}\, H_4}$$

This time it is obtained directly, without Fourier asymptotics. It also shows why $y = y' = z$.

1 The expression and the conventions used

Your expression:

$$\frac{d^3N_{\mathrm{NLO}}}{d^3k} = \frac{4}{(2\pi)^3}\int_\Lambda^{\vee - k^+}\frac{dp^+}{2\pi}\int_{z,w,w'} e^{-ik(w'-w)}\int_{y,y'} \times\Big[\mathfrak{A}^{\dagger(1)e'}_{k'}(P^+,-y')U^{\dagger e'c'}(y')U^{d'b}(z)\bar{\mathfrak{C}}^{\dagger d'ac'}_{1jik'} - \bar{\mathfrak{A}}^{\dagger(1)c'}_{k'}(P^+,-y')\bar{\mathfrak{C}}^{\dagger d'ac'}_{1jik'}U^{d'b}(z)\Big] \times\Big[\bar{\mathfrak{C}}^{dac}_{1jik}U^{bd}(z)U^{ce}(y)\mathfrak{A}^{(1)e}_k(P^+,-y) - U^{bd}(z)\bar{\mathfrak{C}}^{dac}_{1jik}\bar{\mathfrak{A}}^{(1)c}_k(P^+,-y)\Big], \quad (1.1)$$

with $P^+ \equiv p^+ + k^+$, $\bar{\mathfrak{C}}_{1jik} \equiv \bar{\mathfrak{C}}_{1jik}(P^+, y-w; p^+, w-z)$, and the conjugate one at $(y'-w'; w'-z)$.

What each index is.

- $\mathfrak{C}^{bca}_{1jki}(k^+, z; p^+, x)$ is the splitting of a parent (momentum k^+, polarization i) into a gluon of momentum p^+ (polarization j) and a gluon of momentum $k^+ - p^+$ (polarization k). The $\frac{k^+}{p^+}$ term has j along x, so j is the p^+ gluon.
- In $\bar{\mathfrak{C}}_{1jik}(P^+, y-w; p^+, w-z)$ the slots are relabelled:
  - k is the parent: momentum P^+, position y, colour c, contracted with $\mathfrak{A}_k$;
  - j is the unmeasured gluon: momentum p^+, position z, colour d;
  - i is the measured gluon: momentum k^+, position w, colour a.

So, with $x \equiv w - z$, the tensor of $\bar{\mathfrak{C}}_{1jik}$ is

$$T^m_{ijk} = \delta_{km}\delta_{ij} - \alpha\,\delta_{jm}\delta_{ik} - \beta\,\delta_{im}\delta_{jk}, \qquad \alpha = \frac{P^+}{p^+}, \quad \beta = \frac{P^+}{k^+}. \quad (1.2)$$

Two things to check in the expression.

1. In the conjugate bracket $U^{d'b}(z)$ should be $U^{\dagger d'b}(z) = U^{bd'}(z)$. Then $\sum_b U^{bd'}(z)U^{bd}(z) = \delta^{dd'}$: the Wilson lines of the unmeasured gluon cancel, as they must for an integrated-out gluon. I use this.
2. $\bar{\mathfrak{A}}^c_k(P^+, -y)$ is $\mathfrak{A}$ with $U(x)\rho(x)$: $\bar{\mathfrak{A}}^c_k \propto \int_x \mathcal{K}^k(x-y)U^{ce}(x)\rho^e(x)$. Then the amplitude bracket is

$$\bar{\mathfrak{C}}^{dac}_{1jik}U^{bd}(z)\,\frac{ig}{\sqrt{2}\pi\sqrt{P^+}}\,L^c_k(y), \qquad L^c_k(y) \equiv \int_x \mathcal{K}^k(x-y)\big[U^{ce}(y) - U^{ce}(x)\big]\rho^e(x), \quad (1.3)$$

which is the LO amplitude of the parent gluon at y. The overall sign convention of $\mathfrak{A}$ cancels between amplitude and conjugate.

2 Step 1: the δ-function puts the parent at $y = \xi w + (1-\xi)z$

$\mathfrak{C}(k^+, z; p^+, x) \propto \delta^{(2)}\big(z + \frac{p^+}{k^+}x\big)$. In $\bar{\mathfrak{C}}(P^+, y-w; p^+, w-z)$ this is

$$\delta^{(2)}\Big(y - w + \frac{p^+}{P^+}(w-z)\Big) \quad \Rightarrow \quad y = \frac{k^+}{P^+}\,w + \frac{p^+}{P^+}\,z = \xi\,w + (1-\xi)\,z, \qquad \xi \equiv \frac{k^+}{p^+ + k^+}. \tag{2.1}$$

The parent sits at the $+$-momentum-weighted centre of its two daughters. Likewise $y' = \xi w' + (1-\xi)z$. The y, y' integrals are done by these δ's. Hence

$$y = z + \xi(w-z), \qquad y' = z + \xi(w'-z), \qquad y' - y = \xi(w'-w). \tag{2.2}$$

3 Step 2: colour, normalization and the p^+ integral

Colour. With $\delta^{dd'}$ from Section 1, $\sum_{a,d} f^{c'da} f^{cda} = N_c\,\delta^{cc'}$. So amplitude and conjugate have the same parent colour c, contracted between $L^c_k(y)$ and $L^c_{k'}(y')^\dagger$.

Factors.

- $|\mathfrak{C}|^2 \to \frac{g^2 N_c}{4\pi^2\,2P^+}\,\frac{p^+k^+}{P^{+2}}\,\frac{x^m x'^{m'}}{x^2 x'^2}\,T^m_{ijk}T^{m'}_{ijk'}$, with $x = w - z$, $x' = w' - z$.
- $|\mathfrak{A}|^2 \to \frac{g^2}{2\pi^2 P^+} L_k L'_{k'}$.
- $\frac{p^+k^+}{P^{+2}} = \xi(1-\xi)$.

Change of variable. $p^+ = k^+\frac{1-\xi}{\xi}$, $dp^+ = -\frac{k^+}{\xi^2}d\xi$, so $\int_\Lambda^{\vee - k^+}\frac{dp^+}{P^{+2}} = \frac{1}{k^+}\int_a^{\xi_{\max}} d\xi$, with $a = \frac{k^+}{\vee}$ and $\xi_{\max} = \frac{k^+}{k^+ + \Lambda}$.

Normalization. $\frac{4}{(2\pi)^3}\cdot\frac{1}{2\pi k^+}\cdot\frac{g^4 N_c}{16\pi^4} = \frac{1}{(2\pi)^3}\frac{g^4 N_c}{8\pi^5 k^+} \equiv \mathcal{N}$.

4 Step 3: the polarization sum (any $d_\perp$)

$$\sum_{i,j} T^m_{ijk}T^{m'}_{ijk'} = (\alpha^2+\beta^2)\,\delta_{mm'}\delta_{kk'} + 2\alpha\beta\,\delta_{mk'}\delta_{km'} + (d_\perp - 2\alpha - 2\beta)\,\delta_{km}\delta_{k'm'}. \tag{4.1}$$

Multiply by $\xi(1-\xi)$, using $\alpha = \frac{1}{1-\xi}$ and $\beta = \frac{1}{\xi}$:

$$n_1 = \underbrace{\frac{1-\xi}{\xi}}_{\beta^2\xi(1-\xi)} + \frac{\xi}{1-\xi}, \qquad n_2 = 2, \qquad n_3 = d_\perp\,\xi(1-\xi) - 2. \tag{4.2}$$

(Checked for $d = 2, 3, 4$ and symbolically: `from_amplitude/checks.py`.)

5 Step 4: Row IV, exact

$$\begin{aligned}\text{Row IV} = \mathcal{N}\int_a^{\xi_{\max}} d\xi \int_{z,w,w'} e^{-ik(w'-w)}\Big[&n_1\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;L(y)\cdot L^\dagger(y') \\ &+ n_2\,\big(\mathcal{K}(w-z)\cdot L^\dagger(y')\big)\big(\mathcal{K}(w'-z)\cdot L(y)\big) + n_3\,\big(\mathcal{K}(w-z)\cdot L(y)\big)\big(\mathcal{K}(w'-z)\cdot L^\dagger(y')\big)\Big],\end{aligned} \tag{5.1}$$

with y, y' from Eq. (2.2) and the colour c summed in each product of L's.

- $\vee$ appears only in the lower limit $a = \frac{k^+}{\vee}$.
- There is no explicit $\log(\frac{\vee}{k^+} - 1)$ here. The one in line 2 of the three-line form is produced by the rewriting into a plus prescription and cancels again (see `RowIV_BFKL_only.pdf`).

6 Step 5: the $\log \frac{\vee}{k^+}$

In Eq. (5.1) the only ξ-dependence, apart from $n_{1,2,3}$, is through $y = z + \xi(w - z)$ and $y' = z + \xi(w' - z)$. As $\xi \to 0$ these go smoothly to $y, y' \to z$.

- $n_2 = 2$ and $n_3 = d_\perp \xi(1-\xi) - 2$ are finite at $\xi = 0$. So their integrals have no $\log \frac{1}{a}$.
- $\frac{\xi}{1-\xi}$ in n_1 is finite at $\xi = 0$ (it is the $\xi \to 1$, i.e. JIMWLK, end).
- **Only** $\beta^2 \xi(1-\xi) = \frac{1-\xi}{\xi}$ **is singular**, and $\int_a \frac{d\xi}{\xi} = \log \frac{\vee}{k^+} + \dots$.

So the coefficient of $\log \frac{\vee}{k^+}$ is the n_1 integrand at $\xi = 0$, i.e. at $y = y' = z$:

$$\text{Row IV} = \log \frac{\vee}{k^+} \, \mathcal{N} \int_{z,w,w'} e^{-ik(w'-w)} \, \mathcal{K}(w-z) \cdot \mathcal{K}(w'-z) \, L(z) \cdot L^\dagger(z) \; + \; \big(\text{no } \log \vee\big). \tag{6.1}$$

- At $\xi = 0$ the w, w' integrals are finite: $\int_{w,w'} e^{-ik(w'-w)} \mathcal{K}(w-z) \cdot \mathcal{K}(w'-z) = \frac{4\pi^2}{\mathbf{k}^2}$, with no $\log \mu$ and no pole.
- At finite ξ Eq. (5.1) has the collinear pole. Its coefficient, however, is the Fourier transform of a smooth function at momentum $\mathbf{k}/\xi$. It vanishes faster than any power as $\xi \to 0$, so it does not touch the $\log \vee$.

Why $y = y' = z$. This answers your question. y and y' are the parent's positions in the amplitude and in the conjugate. By Eq. (2.2), $y = \xi w + (1-\xi) z$: the parent is at the $+$-momentum-weighted centre of its daughters. In the BFKL region $\xi \to 0$ ($p^+ \gg k^+$), the unmeasured gluon at z carries all the $+$ momentum. So the parent sits at z, in the amplitude and in the conjugate alike. The same point z was found earlier from the Fourier transform in the (y, y') variables; here it is just kinematics.

7 Result

Write out $L(z) \cdot L^\dagger(z)$:

$$\boxed{\begin{aligned} \text{BFKL}\big|_{\text{II,IV}} &= \log \frac{\vee}{k^+} \, H_4, \\ H_4 &= \frac{1}{(2\pi)^3} \frac{g^4 N_c}{8\pi^5} \frac{1}{k^+} \int_{x,x',z,w,w'} e^{-ik(w'-w)} \, \mathcal{K}(w-z) \cdot \mathcal{K}(w'-z) \, \mathcal{K}(x-z) \cdot \mathcal{K}(x'-z) \\ &\quad \times \big[U(z) - U(x')\big]^{e'c} \big[U(z) - U(x)\big]^{ce} \, \rho^{e'}(x') \rho^e(x). \end{aligned}} \tag{7.1}$$

- This is exactly the H_4 obtained before from the three-line form, with the same prefactor $\frac{g^4 N_c}{8\pi^5}$, the same kernels and the same Wilson lines.
- Expanded, the bracket gives your four lines: $U(x')U(x)\rho\rho$, $\rho\rho$ (from $U(z)U^\dagger(z) = 1$), $-U(z)U(x)\rho\rho$, $-U(x')U(z)\rho\rho$.
- The colour orientation follows your $\mathfrak{A}$, $\bar{\mathfrak{A}}$; in your LO notation the first factor reads $U^{e'c}(y')$ and the second $U^{ec}(y)$.

Physical picture. The term comes from $\beta^2 = (P^+/k^+)^2$ in $|\mathfrak{C}|^2$. That is the piece where the *measured* gluon's polarization is along $x = w - z$ (δ_{im}) and the parent's polarization passes to the fast gluon (δ_{jk}). It is the soft emission of the measured gluon, with kernel $\mathcal{K}(w-z)$, from a fast gluon produced at z with the LO amplitude $L(z)$. Its weight $\frac{1-\xi}{\xi}$ is the $\frac{1}{\xi}$ pole of P_{gg}.

8 Comparison with your three-line form

To compare, go to the variables (y, y'): by Eq. (2.2), $d^2w \, d^2w' = \xi^{-4} d^2y \, d^2y'$ and $\frac{x^m}{x^2} = \xi \frac{(y-z)^m}{(y-z)^2}$. This gives the $\frac{d\xi}{\xi^2}$ and the phase $e^{-ik(y'-y)/\xi}$. The z integral is then

$$\int_z \frac{(y-z)^m (y'-z)^{m'}}{(y-z)^2 (y'-z)^2} = A \, \delta_{mm'} + B' \, \frac{X^m X^{m'}}{X^2}, \qquad B' = (d_\perp - 2) A \to -\pi. \tag{8.1}$$

A carries the pole and the $\ln X^2$; the finite $B' = -\pi$ is checked numerically. Then:

- The $L \cdot L'$ coefficient is $n_1(2d_\perp - 2)A + d_\perp \xi(1-\xi) A = d_\perp A\big[\xi(1-\xi) + \frac{1-\xi}{\xi} + \frac{\xi}{1-\xi}\big] + (d_\perp - 2) A \, n_1$. This is your bracket in line 1.

- The $\frac{(X\cdot L)(X\cdot L')}{X^2}$ coefficient is $d_\perp \xi(1-\xi)B'/\xi^2 = -\pi\, d_\perp \frac{1-\xi}{\xi}$. This is your $-d_\perp \frac{1-\xi}{\xi}$ term in line 3.
- The $\frac{\xi}{1-\xi}$ term with the upper limit $\xi_{\max}$ gives your $\log \frac{k^+}{\Lambda}$ line.

So the ξ-structure of your three lines is reproduced. I did not recheck the ϵ-constants of A here; they do not enter the BFKL term. Both routes give the same $\log \frac{\vee}{k^+} H_4$.

Checks (`from_amplitude/checks.py`):

- polarization sum, exact for $d = 2, 3, 4$;
- $n_{1,2,3}$, symbolic;
- $f^{c'da} f^{cda} = N_c \delta^{cc'}$ for SU(3), to 10^{-15};
- normalization $\mathcal{N}$, symbolic;
- $B' = -3.14159265359$.